

MATH 373: Intro to Machine Learning

Case Study: Tree Based Methods

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Directions: Provide all code and written answers for the following questions.

1. First, go through these two brief Python tutorials on Trees here: <https://www.datacamp.com/tutorial/decision-tree-classification-python> and n Random Forests here: <https://data36.com/random-forest-in-python/>. Make sure your code works and you are able to re-implement the code there. **You do not have to present this in the homework, I just want you to have a framework for implementing trees and random forests in Python.**
2. Now we will revisit the *Mashable* data.
 - (a) Build classifiers for this data set using the tree-based methods that we've learned in class. In particular, build classifiers on the training data and assess the performance of each of the following methods on the test data (available on Canvas):
 - i. Classification tree
 - ii. Bagging
 - iii. Random ForestsFor each method, consider using cross-validation and grid searches to identify the best tuning parameters (like the depth of the tree(s), the number of variables to use in random forests, etc.). The goal here is to build the best classifier that you can for predicting the popularity of a new website.
 - (b) Using variable importance techniques with random forests, provide a plot of which variables in this data set are most important for your final predictive model.
 - (c) What is the MSPE for each of your fitted models? Compare and contrast between models and then compare and contrast these ensemble-based models with the classification models that you fit in your last homework. What are the advantages and disadvantages of using these ensemble methods?
3. Run the above tree-based methods above on the SPAM email data from the Naïve Bayes case study, making sure again to build the best model possible. Compare the performance of your methods with the results of your Naïve Bayes classifier from before and identify what key words or phrases were most important for your SPAM prediction using variable importance techniques.